AI MCP Server: Test Report

Generic ERPNext MCP server (**frappe**) for the AI-Assisted Activity Tracker and Controller. Tested end to end through Telegram Web, 2026-10-22, branch **ai-mcp-erpnext**.

1. Summary

The Telegram assistant had built-in tools for sales only (leads, opportunities, quotations, sales orders, activities). The **frappe** MCP server is a second toolbox that reaches any other ERPNext DocType (stock, accounting, projects, notes, users read-only). It runs **as the calling user**, so ERPNext's own permissions apply, and every change waits for the user to reply **yes**.

Outcome: reads, the yes-gate, the no-gate and per-user permission enforcement all worked through Telegram.

2. How it works

Step	What happens
1	The user chats with the Telegram bot; the bot maps the Telegram id to an ERPNext caller (here: **manager**).
2	The API runs the LLM with the **built-in sales tools first** (phase 1).
3	Only if the model says it cannot do the request (or runs out of steps), the MCP tools are added (phase 2).
4	**Read tools** run at once. **Write tools** are parked; the user sees a bullet preview and must reply yes or no.
5	On yes, the API calls the MCP server with the caller's name in _**meta.caller**; the server uses that person's own ERPNext API key. ERPNext allows or denies.

Tool	Type	Purpose
list_docs, get_doc, count_docs, describe_doctype, run_report	read	List, open, count, inspect fields, run saved reports
create_doc, update_doc, submit_doc	write (needs yes)	Create, change, submit records
cancel_doc	write, destructive (needs yes)	Cancel a submitted record

Never available: delete, arbitrary method calls, writes to User/Role/DocPerm/API keys/System Settings (User is readable). Sales Order submit/cancel only for CONFIRM_CALLERS; Item create only for ITEM_CREATORS.

3. Environment

Item	Value
MCP server	erpnext/mcp_server/frappe_server.py, stdio, JSON-RPC 2.0 (MCP 2024-11-05), 9 tools
Identity	Telegram user 1722472434 maps to caller manager (Sales Manager + Item Manager)
ERPNext	v15.124.0 (Frappe 15.123.1), local
LLM	Groq openai/gpt-oss-120b (fallback gpt-oss-20b)
Bot / API	telegram_bot.py (long polling); FastAPI on port 8100
Offline test suite	123 passed

4. Round 1 results (Telegram)

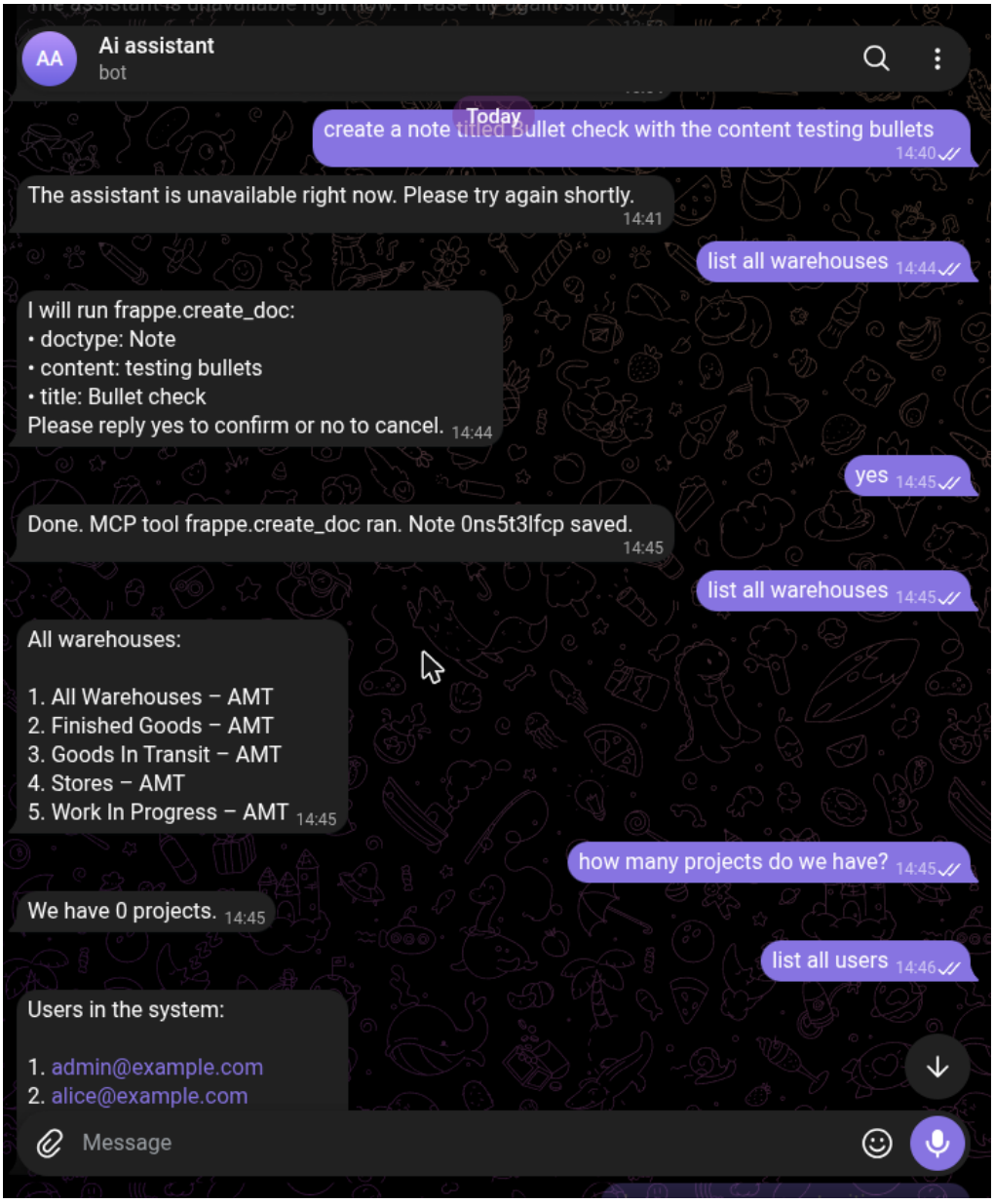
Prompt	Result
list all warehouses	Pass: 5 warehouses listed
how many employees do we have in the system?	Pass: ERPNext denied manager access to Employee, shown to the user
list all customer groups	Pass: 5 groups listed
create a customer group called Test Group MCP	Preview parked; after yes ERPNext denied manager. Nothing created (permission enforced)
create a note titled MCP test note...	Pass: preview, nothing written before yes; after yes Note created, owner manager
create a note titled Cancel me..., then no	Pass: Cancelled, nothing written
how many projects / list all users and their roles	Blocked by Groq 429 rate limits; verified at server level instead

Direct server checks (no LLM, real ERPNext)

Call	manager	alice	bob
count_docs Warehouse / Customer	5 / 6	5 / 6	5 / 6
list_docs User (after the read-only change)	4 users	4 users	4 users
update_doc User	blocked	blocked	blocked
cancel_doc Sales Order	passes guard	blocked	blocked
any call with no caller	refused	refused	refused

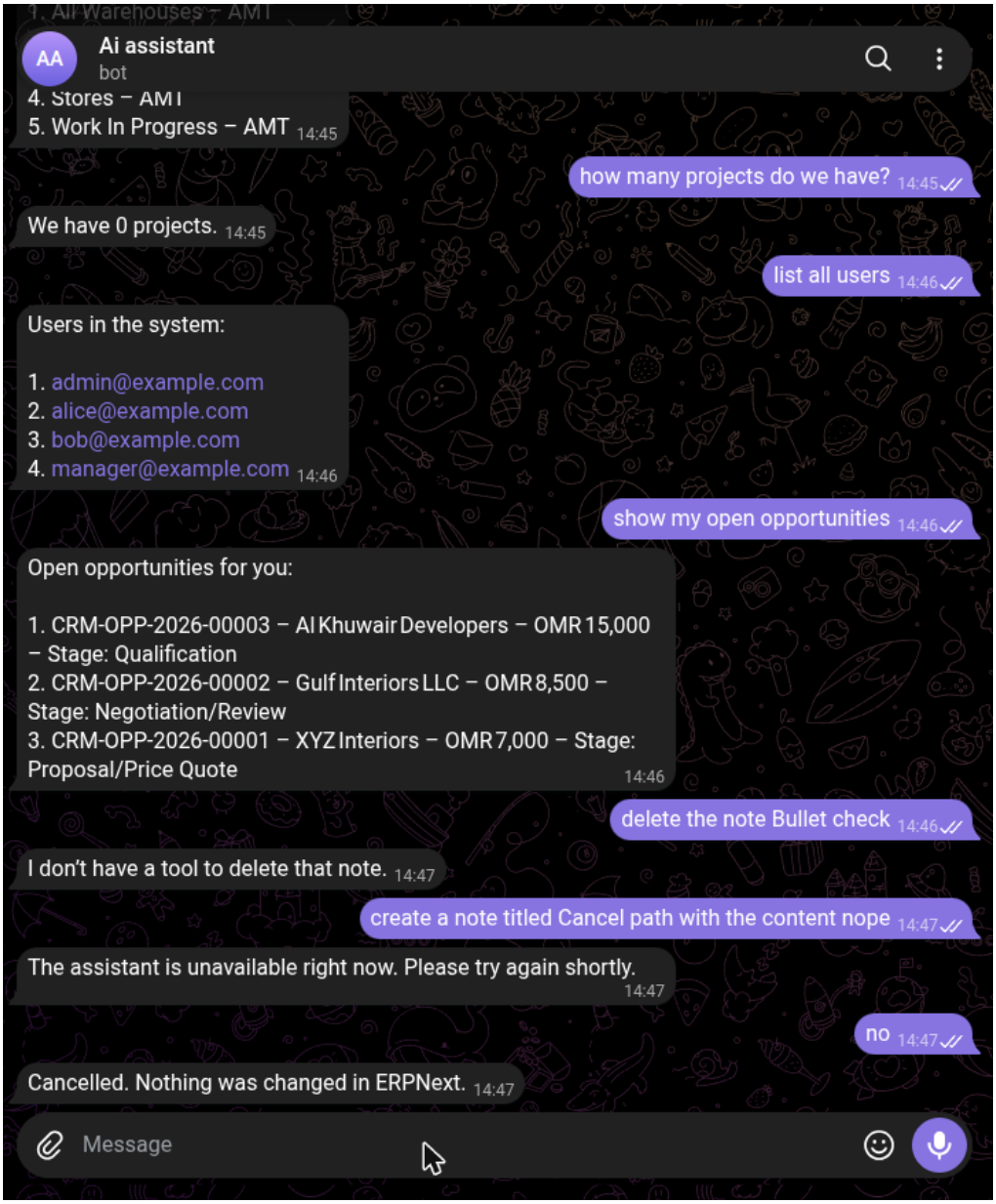
5. Round 2: workflow screenshots

Chat-only captures of the Telegram conversation with the bot, in order.



Screen 1. Write preview, then yes, then reads.

Prompt	Hits MCP?	Result
create a note titled Bullet check...	yes, create_doc (write)	Bullet preview, nothing written yet
yes	yes	Note 0ns5t3lfcp saved
list all warehouses	yes, list_docs (read)	Numbered list of the 5 warehouses
how many projects do we have?	yes, count_docs (read)	We have 0 projects.
list all users	yes, list_docs on User (read-only)	The four accounts

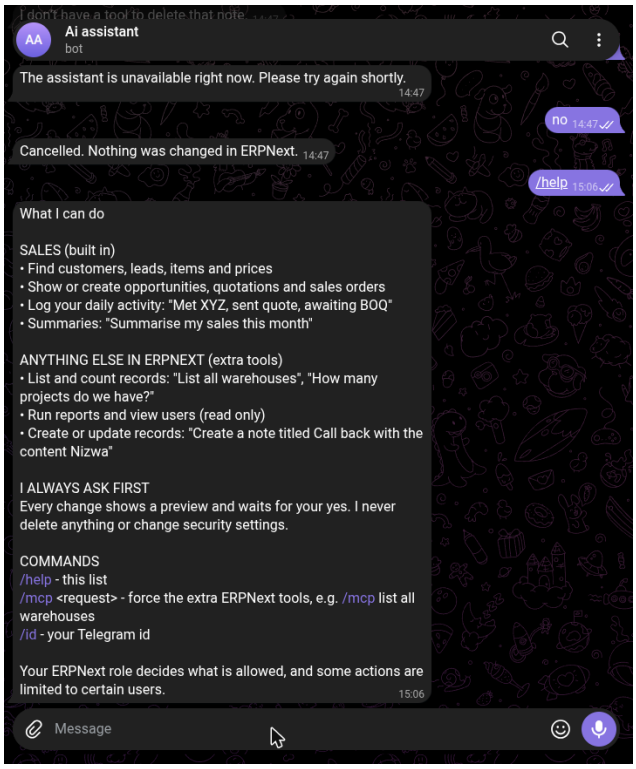


Screen 2. Built-in control, delete refusal and the no path.

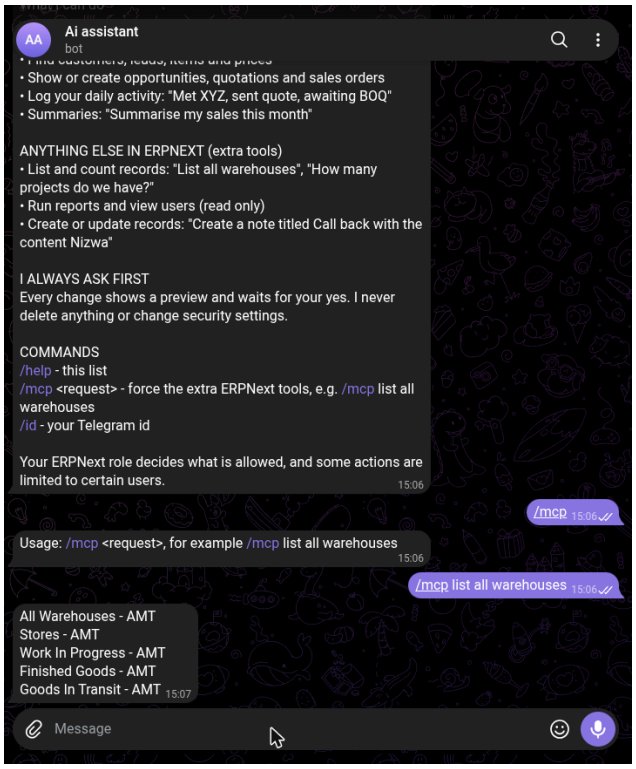
Prompt	Hits MCP?	Result
show my open opportunities	no, built-in search_opportunities	3 open opportunities
delete the note Bullet check	no (no delete tool)	I don't have a tool to delete that note.
create a note titled Cancel path..., then no	yes (write)	Cancelled. Nothing was changed in ERPNext. The Note does not exist.

6. Round 3: /help and /mcp

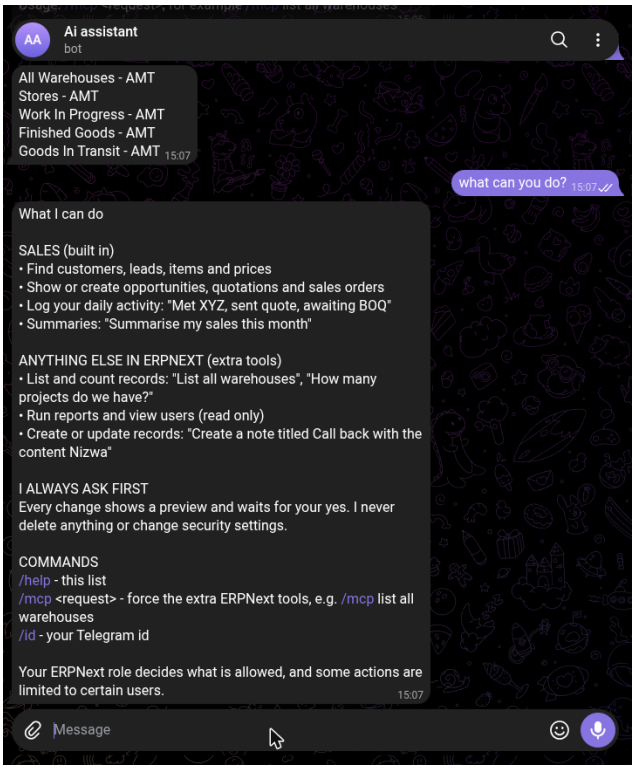
New bot commands. /help and the plain question are answered by the bot itself (no AI call, so instant); /mcp sends the request with force_mcp so the built-in sales tools are skipped and only the MCP tools are offered.



/help lists what the bot can do, what it always asks first, and the commands.



/mcp with no request shows the usage; **/mcp list all warehouses** uses the MCP tool directly.



A plain **what can you do?** gets the same fixed help.

Message	Handled by	Result
/help, /start, help, what can you do?	the bot (no middleware call)	Fixed help text
/mcp	the bot	Usage: /mcp <request>, for example /mcp list all warehouses
/mcp list all warehouses	API with force_mcp=true, MCP only	The 5 warehouses (list_docs)

7. Prompt catalogue (docs/prompts.csv)

54 prompts classified: **29** hit the MCP server, **25** do not (built-in sales tools, no tool, refused, or answered in code). Rows marked 'yes (gated)' reach MCP but are restricted by CONFIRM_CALLERS.

#	Prompt	Hits MCP	Handled by	R/W	Yes?
1	Show my open opportunities	no	search_opportunities	read	no
2	What quotations did we send last month?	no	list_quotations	read	no
3	Show sales orders finalized this week	no	list_sales_orders	read	no
4	Summarise my sales this month	no	sales_summary	read	no
5	Find customer Nizwa Builders	no	search_clients	read	no
6	Find item workstation	no	search_items	read	no
7	Met XYZ Interiors today and sent a quotation awaiting BOQ	no	propose_log_activity	write	yes
8	Create a quotation for Nizwa Builders for 5 workstations	no	propose_create_quotation	write	yes
9	Convert quotation SAL-QTN-2026-00004 to a sales order	no	propose_quotation_to_order	write	yes
10	Create an opportunity for XYZ Interiors worth 5000	no	propose_create_opportunity	write	yes
11	List all warehouses	yes	list_docs	read	no
12	How many warehouses do we have?	yes	count_docs	read	no
13	Show the stock balance report	yes	run_report	read	no
14	Create a warehouse called Test WH	yes	create_doc	write	yes
15	List all item groups	yes	list_docs	read	no
16	Which items are in the Stores warehouse?	yes	list_docs (Bin)	read	no
17	List all customer groups	yes	list_docs	read	no
18	Create a customer group called Test Group MCP	yes	create_doc	write	yes
19	List all territories	yes	list_docs	read	no
20	How many suppliers do we have?	yes	count_docs	read	no
21	How many employees do we have?	yes	count_docs	read	no
22	List all departments	yes	list_docs	read	no
23	Show leave applications	yes	list_docs	read	no
24	How many projects do we have?	yes	count_docs	read	no
25	List all tasks	yes	list_docs	read	no
26	Create a project called Website Revamp	yes	create_doc	write	yes
27	Show the general ledger report	yes	run_report	read	no
28	List all price lists	yes	list_docs	read	no
29	List unpaid sales invoices	yes	list_docs	read	no
30	Create a note titled MCP test note with the content hello	yes	create_doc	write	yes
31	Create a note titled Cancel me then reply no	yes	create_doc	write	yes
32	Add a todo to call Nizwa Builders tomorrow	yes	create_doc (ToDo)	write	yes
33	List my notes	yes	list_docs	read	no
34	List all users	yes	list_docs (User)	read	no

#	Prompt	Hits MCP	Handled by	R/W	Yes?
35	List all users and their roles	yes	list_docs / get_doc	read	no
36	Change the role of bob to System Manager	no (refused)	none	write	no
37	Create a new user called Sam	no (refused)	none	write	no
38	Delete the note MCP test note	no (refused)	none	write	no
39	Show system settings	no (refused)	none	read	no
40	List all API keys	no (refused)	none	read	no
41	Cancel sales order SAL-ORD-2026-00004	yes (gated)	cancel_doc	write	yes
42	Submit sales order SAL-ORD-2026-00003	yes (gated)	submit_doc	write	yes
43	Create an item called Test Chair priced 50	no	propose_create_item	write	yes
44	What time is it in Muscat?	no	none	read	no
45	Fetch https://example.com and summarise it	no	none	read	no
46	Hello	no	none	read	no
47	Thanks	no	none	read	no
48	yes (with nothing pending)	no	none	n/a	n/a
49	no (with something pending)	no	executor	n/a	n/a
50	/help	no	bot (fixed text)	n/a	no
51	what can you do?	no	bot (fixed text)	n/a	no
52	/mcp list all warehouses	yes	list_docs (force_mcp)	read	no
53	/mcp	no	bot	n/a	no
54	/mcp show my open opportunities	yes	list_docs (force_mcp)	read	no

8. Caveats

Item	Detail
Groq rate limits	429s on both models blocked some prompts and slowed the rest; an environment limit, not an MCP failure.
Coverage	submit/cancel and run_report/describe_doctype were not exercised live. Only manager is linked to Telegram; alice/bob were checked at server level.
Safety scope	The built-in tools' fuzzy-name and price-from-ERPNext rules do not apply to MCP requests; ERPNext validation and the caller's roles are the backstop.
Test data	Notes created during testing were deleted afterwards.